Voice2Gov

Database Design Software Requirements Specification

Version 1.0

Project	Voice2Gov
Document Type	Database Design SRS
Database	MongoDB
Backend	Python + FastAPI
Frontend	React Native
Database Architecture	MongoDB Document Database
Primary Users	Citizens, Organization Staff, Organization Administrators, System Administrators

Prepared for project development, review, academic submission, and stakeholder sharing.

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1. Introduction

This document defines the database-specific requirements for Voice2Gov. It is derived from the existing Voice2Gov Software Requirements Specification and isolates the MongoDB persistence design. The database supports citizen complaint registration, organization routing, department assignment, community likes, complaint progress tracking, notifications, administration, analytics, and auditability.

2. Database Objectives

• Store authenticated users and role information securely.

• Store organizations, departments, and complaint categories.

• Store complete civic complaint records including location and priority information.

• Store complaint photographs and evidence metadata.

• Prevent duplicate likes from the same user.

• Preserve complaint status and progress history.

• Deliver and track in-app and push notification data.

• Record administrative and organization actions through audit logs.

• Support geospatial complaint queries using MongoDB 2dsphere indexing.

• Support indexed, paginated access for large complaint and notification datasets.

3. Database Scope

The database scope includes the 12 required MongoDB collections: users, organizations, departments, categories, complaints, complaint_images, complaint_likes, complaint_updates, notifications, device_tokens, audit_logs, and admin_settings. The original SRS identifies these collections as the recommended MongoDB design. ■filecite■turn0file0■L767-L774■

4. Database Technology

Layer	Technology	Purpose
Database	MongoDB	Primary persistence layer
Backend	Python + FastAPI	REST API, validation, business logic and database access
Data Format	JSON / BSON	API and document representation
Authentication	JWT + RBAC	Secure identity and authorization
Image Storage	Secure Object Storage / GridFS	Complaint and progress photographs
Location	GeoJSON Point	Complaint geospatial information

5. Database Architecture

Voice2Gov uses a document-oriented MongoDB database accessed by the Python FastAPI backend. Relationships are represented primarily through ObjectId references rather than relational SQL foreign keys. Complaint locations use GeoJSON Point format, and large image binaries should be kept in secure object storage or GridFS while MongoDB stores references and metadata. ■filecite■turn0file0■L255-L268■ ■filecite■turn0file0■L860-L862■

6. Collection Catalogue

Collection	Purpose
users	All authenticated citizens, organization administrators, organization staff and system administrators.
organizations	Government/civic organizations responsible for complaints.
departments	Departments belonging to organizations.
categories	Complaint category master data.
complaints	Core civic complaint records.
complaint_images	Complaint, progress and completion image metadata.
complaint_likes	Individual complaint upvotes/likes.
complaint_updates	Complaint status and progress history.
notifications	In-app notification records.
device_tokens	Push notification device registrations.
audit_logs	Security and administrative audit records.
admin_settings	Platform-level configurable settings.

7. Detailed Collection Specifications

users

Field	Type	Required	Description
_id	ObjectId	Yes	Unique user identifier
name	string	Yes	Full name
email	string	Yes / Unique	Login email
phone	string	No / Unique	Mobile number
password_hash	string	Yes	Argon2/bcrypt hash; never plaintext
role	string	Yes	admin, organization_admin, organization_staff, citizen
organization_id	ObjectId	Conditional	Organization reference for organization users
department_id	ObjectId	Conditional	Department reference for staff
status	string	Yes	active, inactive, suspended, pending
profile_image	string	No	Secure profile image reference
is_email_verified	boolean	No	Email verification state
is_phone_verified	boolean	No	Phone verification state
last_login_at	date	No	Last successful login
created_at	date	Yes	Creation timestamp
updated_at	date	Yes	Last modification timestamp

organizations

Field	Type	Required	Description
_id	ObjectId	Yes	Unique organization identifier
name	string	Yes	Organization name
code	string	Yes / Unique	Organization code
description	string	No	Description
contact_email	string	No	Contact email
contact_phone	string	No	Contact phone
address	string	No	Organization address
status	string	Yes	active or inactive
created_at	date	Yes	Creation timestamp
updated_at	date	Yes	Last modification timestamp

departments

Field	Type	Required	Description
_id	ObjectId	Yes	Unique department identifier
organization_id	ObjectId	Yes	Reference to organizations
name	string	Yes	Department name
code	string	Yes	Department code within organization
description	string	No	Department description

Field	Type	Required	Description
status	string	Yes	active or inactive
created_at	date	Yes	Creation timestamp
updated_at	date	No	Last modification timestamp

categories

Field	Type	Required	Description
_id	ObjectId	Yes	Unique category identifier
name	string	Yes	Category name
code	string	Yes / Unique	Category code
description	string	No	Category description
organization_id	ObjectId	No	Optional organization-specific category
default_department_id	ObjectId	No	Default routing department
priority_weight	number	No	Default priority contribution
status	string	Yes	active or inactive
created_at	date	Yes	Creation timestamp
updated_at	date	Yes	Last modification timestamp

complaints

Field	Type	Required	Description
_id	ObjectId	Yes	Unique complaint identifier
complaint_number	string	Yes / Unique	Human-readable complaint ID
user_id	ObjectId	Yes	Citizen who submitted complaint
organization_id	ObjectId	Yes	Routed organization
department_id	ObjectId	Yes	Routed department
category_id	ObjectId	Yes	Complaint category
title	string	Yes	Complaint title
description	string	Yes	Complaint description
status	string	Yes	SUBMITTED, ONGOING, PROCESSING, COMPLETED, REJECTED, DUPLICATE, CLOSED
visibility	string	Yes	public or private
location	object	Yes	GeoJSON Point: type + [longitude, latitude] + address
likes_count	integer	Yes	Denormalized community like count
priority_score	number	Yes	Overall complaint priority
severity_score	number	No	Severity contribution
community_engagement_score	number	No	Community contribution
age_score	number	No	Age contribution
assigned_to	ObjectId	No	Assigned organization user
assigned_at	date	No	Assignment timestamp
rejection_reason	string	No	Reason for rejection

Field	Type	Required	Description
duplicate_of	ObjectId	No	Reference to duplicate complaint
completion_notes	string	No	Resolution notes
completed_at	date	No	Completion timestamp
created_at	date	Yes	Submission timestamp
updated_at	date	Yes	Last modification timestamp

complaint_images

Field	Type	Required	Description
_id	ObjectId	Yes	Unique image metadata identifier
complaint_id	ObjectId	Yes	Complaint reference
uploaded_by	ObjectId	Yes	User who uploaded image
file_name	string	Yes	Original/sanitized filename
storage_key	string	Yes	Secure object storage key or GridFS reference
file_url	string	No	Accessible secure URL where applicable
mime_type	string	Yes	Validated MIME type
file_size	integer	Yes	File size in bytes
image_type	string	Yes	complaint_evidence, progress_evidence, completion_evidence
caption	string	No	Optional caption
created_at	date	Yes	Upload timestamp

complaint_likes

Field	Type	Required	Description
_id	ObjectId	Yes	Unique like identifier
complaint_id	ObjectId	Yes	Complaint reference
user_id	ObjectId	Yes	User reference
created_at	date	Yes	Like timestamp

complaint_updates

Field	Type	Required	Description
_id	ObjectId	Yes	Unique update identifier
complaint_id	ObjectId	Yes	Complaint reference
organization_id	ObjectId	Yes	Organization reference
department_id	ObjectId	No	Department reference
updated_by	ObjectId	Yes	User who made update
previous_status	string	No	Previous complaint status
new_status	string	No	New complaint status
message	string	No	Progress/update message
images	array<ObjectId>	No	Related evidence image references
created_at	date	Yes	Update timestamp

notifications

Field	Type	Required	Description
_id	ObjectId	Yes	Unique notification identifier
user_id	ObjectId	Yes	Recipient user
type	string	Yes	Notification type
title	string	Yes	Notification title
message	string	Yes	Notification message
complaint_id	ObjectId	No	Related complaint
is_read	boolean	Yes	Read state
read_at	date	No	Read timestamp
created_at	date	Yes	Creation timestamp

device_tokens

Field	Type	Required	Description
_id	ObjectId	Yes	Unique device-token record
user_id	ObjectId	Yes	Device owner
token	string	Yes / Unique	Push notification token
platform	string	Yes	android, ios, web
device_id	string	No	Device identifier
is_active	boolean	Yes	Whether token can receive notifications
last_used_at	date	No	Last token usage
created_at	date	Yes	Creation timestamp
updated_at	date	Yes	Last modification timestamp

audit_logs

Field	Type	Required	Description
_id	ObjectId	Yes	Unique audit record
actor_id	ObjectId	Yes	User performing action
actor_role	string	Yes	Actor role
action	string	Yes	Action performed
entity_type	string	Yes	Entity affected
entity_id	ObjectId	No	Affected entity
metadata	object	No	Additional structured audit information
ip_address	string	No	Request IP where appropriate
user_agent	string	No	Client information
created_at	date	Yes	Audit timestamp

admin_settings

Field	Type	Required	Description
_id	ObjectId	Yes	Unique setting identifier

Field	Type	Required	Description
setting_key	string	Yes / Unique	Configuration key
setting_value	mixed	Yes	Configuration value
data_type	string	Yes	string, number, boolean, object or array
description	string	No	Setting description
updated_by	ObjectId	Yes	Administrator who changed setting
created_at	date	Yes	Creation timestamp
updated_at	date	Yes	Last modification timestamp

8. Collection Relationships

Source	Field	Target	Relationship
users	organization_id	organizations._id	Many-to-one
users	department_id	departments._id	Many-to-one
departments	organization_id	organizations._id	Many-to-one
complaints	user_id	users._id	Many-to-one
complaints	organization_id	organizations._id	Many-to-one
complaints	department_id	departments._id	Many-to-one
complaints	category_id	categories._id	Many-to-one
complaints	assigned_to	users._id	Many-to-one
complaint_images	complaint_id	complaints._id	Many-to-one
complaint_likes	complaint_id	complaints._id	Many-to-one
complaint_likes	user_id	users._id	Many-to-one
complaint_updates	complaint_id	complaints._id	Many-to-one
notifications	user_id	users._id	Many-to-one
notifications	complaint_id	complaints._id	Many-to-one
device_tokens	user_id	users._id	Many-to-one
audit_logs	actor_id	users._id	Many-to-one

MongoDB relationships are implemented through ObjectId references and application-level validation rather than SQL foreign-key constraints.

9. Complaint Status Lifecycle

• SUBMITTED — complaint successfully registered.

• ONGOING — organization has acknowledged the complaint and work has started.

• PROCESSING — complaint is actively being investigated or resolved.

• COMPLETED — work has been completed and the complaint is resolved.

• REJECTED — complaint rejected with a reason.

• DUPLICATE — complaint identified as a duplicate.

• CLOSED — complaint lifecycle administratively closed.

• Primary flow: SUBMITTED → ONGOING → PROCESSING → COMPLETED.

The original SRS requires backend validation to prevent invalid transitions such as COMPLETED → SUBMITTED. ■filecite■turn0file0■L304-L326■

10. Complaint Priority Data Model

The priority model combines community engagement, severity, complaint age, and organization-configured priority. The original SRS specifies configurable weighting so the formula can evolve without changing the mobile application. ■filecite■turn0file0■L289-L302■

Recommended stored components are priority_score, community_engagement_score, severity_score and age_score. Organization-specific weights can be maintained in admin_settings or organization configuration.

11. Indexing Requirements

Collection	Index	Purpose
users	email	Fast login and unique email
users	phone	Fast phone lookup
complaints	user_id	Citizen complaint history
complaints	organization_id	Organization isolation
complaints	department_id	Department queue
complaints	status	Status filtering
complaints	category_id	Category filtering
complaints	created_at	Recent complaints
complaints	priority_score	Priority queue
complaints	location: 2dsphere	Nearby complaints
complaint_likes	complaint_id + user_id	Prevent duplicate likes
notifications	user_id + is_read	Unread notification lookup
complaint_updates	complaint_id + created_at	Complaint timeline

These indexing requirements reflect the MongoDB indexing requirements in the original SRS. ■filecite■turn0file0■L806-L835■

12. Data Integrity and Constraints

• User email must be unique.

• Organization code must be unique.

• Complaint number must be unique.

• Complaint likes must use a unique compound index on complaint_id + user_id.

• Complaint status must be restricted to the defined lifecycle values.

• Invalid status transitions must be rejected by the backend.

• Organization users must be restricted to their own organization records.

• Required ObjectId references must be validated before insertion or update.

• Coordinates must follow [longitude, latitude] GeoJSON ordering.

• Timestamps should be stored consistently in UTC.

13. File and Image Storage

Complaint photographs, progress evidence, completion evidence, and profile images should be represented in MongoDB by metadata and secure storage references. The original SRS recommends secure object storage or GridFS. ■filecite■turn0file0■L246-L249■ ■filecite■turn0file0■L860-L862■

14. Security Requirements

• JWT access and refresh token architecture.

• Argon2 or bcrypt password hashing.

• Role-based authorization.

• Organization-level tenant isolation.

• Pydantic/FastAPI input validation.

• MIME type, file-size and filename validation.

• Rate limiting for login, OTP, uploads, likes and password reset.

• HTTPS for application traffic.

• No sensitive citizen data in public feed.

• Audit logs for important administrative and organization actions.

15. Database Query Requirements

• Citizen complaints by user_id with created_at sorting.

• Organization queue by organization_id.

• Department queue by department_id.

• Public community feed with pagination.

• Nearby complaints using the 2dsphere index.

• Complaint timeline by complaint_id ordered by created_at.

• Unread notifications by user_id and is_read.

• High-priority complaints ordered by priority_score.

16. Data Flow

Complaint creation: users → categories → organizations/departments → complaints → complaint_images → complaint_updates → notifications → audit_logs.

Community engagement: users → complaints → complaint_likes → complaint priority.

Organization processing: complaints → users/departments → complaint_updates → complaint_images → notifications → audit_logs.

Push notifications: users → device_tokens → notifications.

17. Backup, Reliability and Performance

• Automated MongoDB backup and recovery procedures.

• Pagination for large lists.

• Indexes for common queries.

• Aggregation pipelines for dashboard analytics.

• Asynchronous FastAPI operations where appropriate.

• Asynchronous processing for heavy image and notification work.

• Retries for transient storage and notification failures.

• Meaningful database/API error handling.

• Target normal API response time below 500 ms where practical.

• Target normal feed loading below 2 seconds under normal network conditions.

These performance targets align with the original SRS. ■filecite■turn0file0■L864-L879■

18. Database Acceptance Criteria

• All 12 required MongoDB collections are created.

• Each collection contains the defined fields and types.

• ObjectId references are used consistently.

• Complaints support user, organization, department, category, location, status, priority and assignment.

• Complaint images use metadata and secure storage references.

• Duplicate likes are prevented.

• Status transitions are validated.

• GeoJSON locations support a 2dsphere index.

• Complaint progress history is preserved.

• Notifications and device tokens support notification delivery.

• Administrative actions are auditable.

• Organization tenant isolation is enforced.

• Required indexes are created and tested.

• Citizen, organization and admin portal queries are supported.

• Backup and recovery procedures are defined before production deployment.

Source Basis

This database-specific SRS is based on the supplied Voice2Gov Software Requirements Specification v1.0, especially its technology stack, architecture, complaint workflow, authentication/RBAC, MongoDB design, indexing, security, and performance requirements. ■filecite■turn0file0■L232-L253■ ■filecite■turn0file0■L767-L835■